On the Optimal Self-Supervised Multi-Fault Detector for Temperature Sensor Data

Latifa Nabila Harfiya[1], Yan-Cheng Hsu[2], Yung-Hui Li[3], and Jia-Ching Wang[1]
[1] Department of Computer Science and Information Engineering, National Central University, Taiwan
[2] Department of Computer Science and Engineering, University of California San Diego, USA
[3] AI Research Center, Hon Hai Research Institute, Taiwan
E-mail: latifanaharfiya@g.ncu.edu.tw, bill.yancheng.hsu@gmail.com, yunghui@foxconn.com, jcw@csie.ncu.edu.tw

***Abstract* — Accurate detection of faults in sensor data is essential for monitoring and controlling industrial processes, environmental conditions, and infrastructure to ensure reliability and enable informed decision-making. Resolving these faults ensures measurement quality and unlocks process optimization opportunities, resulting in improved performance, energy efficiency, and cost savings. We present a transformer-based fault detection model which adopts the anomaly-attention mechanism. Experiments have been performed on the benchmark faults injected Intel temperature sensor datasets using precision, recall, and F1-score metrics. The result outperforms the other classical and complex algorithms, proving our method's effectiveness.**

I. INTRODUCTION

Currently, there is a significant shift happening in the manufacturing industry. Integrating big data and machine learning, especially deep learning, is revolutionizing traditional manufacturing practices, ushering in the era of smart manufacturing known as Industry 4.0 (I4.0). This shift presents new opportunities for manufacturers. Within this context, the Industrial Internet of Things (IIoT) emerges and connects machines through various sensors, RFID tags, software, and electronics, allowing real-time data collection. The proliferation of intelligent sensors and the IoT serves as crucial facilitators for collecting and storing vast amounts of data in industrial settings [1, 2]. Sensor data is often used to detect abnormal conditions or events requiring immediate attention or action. It is crucial for accurately monitoring and controlling industrial processes, environmental conditions, and infrastructure. Detecting faults in sensor data ensures the reliability and quality of the measurements, which is vital for making informed decisions and maintaining process integrity. By identifying and resolving these faults, process optimization opportunities can be realized, leading to improved performance, reduced energy consumption, and cost savings [3, 4].

Sensor data can encompass various types of information, including numerical readings, categorical data, or even event-based data. Time series data refers to data collected and recorded over a sequence of time intervals or timestamps, where the order of the observations is essential. This is commonly seen in sensor data that captures measurements at regular intervals, such as temperature readings, humidity levels, or vibration intensity recorded at specific time points. Many works address the temporal variation of time series data, which is more meaningful and can reveal intrinsic properties of time series, such as continuity, regularity or trends, because one single point in time typically cannot give enough information for analysis [5].

A fault detection framework has been presented in [3] using a Support Vector Machine (SVM), which is optimized with a biologically-inspired technique, Grey Wolf Optimization (GWO). The results suggest that when the failure rate stays at a low level, an accurate diagnosis of sensor fault can be made. Recurrent Neural Network (RNN) is widely considered a powerful tool for analyzing and modeling time series data due to its ability to capture sequential patterns and dependencies. In [3] and [6], Long Short-Term Memory (LSTM), a variant of RNN, is utilized to make an end-to-end prediction fault diagnosis to prognosis. On the other hand, a stacked LSTM has fully exploited the long-term dependencies information and improved the fault diagnosis performance in [7]. Although using LSTM might be sufficient, architectures like Transformers, originally introduced for natural language processing tasks, have also gained attention in the time series domain, especially for tasks involving long sequences or capturing global dependencies. Several research studies [8-10] and practical applications have demonstrated the effectiveness of transformers in time series anomaly detection. Using transformers, one can capture long-range dependencies, exploit global context, and leverage self-attention mechanisms to

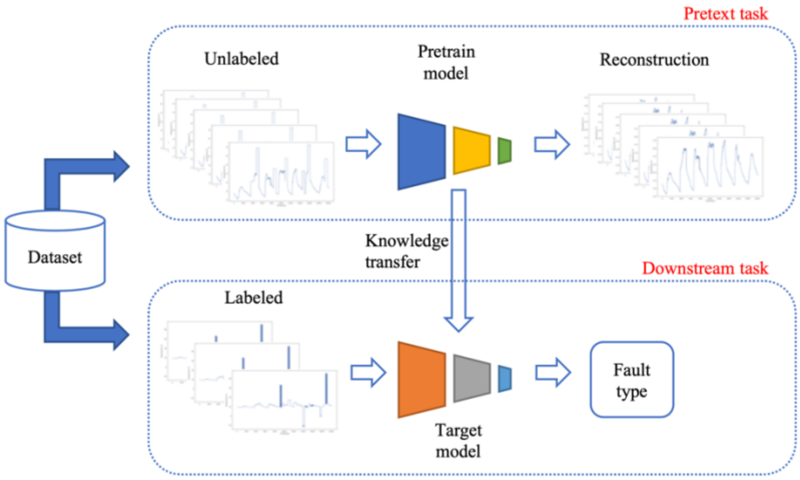


Fig 1. Proposed framework

detects anomalies in complex and high-dimensional time series data.

In order to obtain a progressive detection method to help the model distinguish sensor conditions, we adopt a Transformer-based network architecture from [11]. The literature utilizes the network to solve an unsupervised anomaly detection task. Inspired by a Self-Supervised Learning (SSL) approach, we first train the network model as a denoising network that attempts to remove the faults, which we consider as noise, from the one-dimensional input time series signal and reconstruct the output like the original clean input. In SSL, this first step is known as the pretext task. Next, given less quantity of labeled data, we finetuned the pretrained network model to detect faults based on their class. This step is known as the downstream task or target task in SSL. Furthermore, the experimental results are examined in a comparative manner to confirm the effectiveness of the method presented in this research paper for detecting temperature sensor faults.

II. METHOD

The proposed framework, shown in Fig. 1, is divided into two phases: the pretext task to direct the model toward acquiring intermediary representations of data and the downstream task to transfer the knowledge from the finetuned pretext model. We split our dataset for the pretext task and the downstream task. We use the interpolated time-series signal for each task to help the model observe the learnable pattern. We use the clean signal as the ground truth for the first task and the class label as the ground truth for the latter. Then, we apply the preprocessing techniques outlined in [11], including dividing the dataset into consecutive non-overlapping segments using a sliding window approach. The adopted network architecture introduces association discrepancy and incorporates an Anomaly-Attention mechanism that models the prior-association and series-association to the vanilla Transformer as the core element.

A. *Anomaly-Attention Block*

Instead of using point-wise representation, the original self-attention mechanism properties, using prior-association, which benefiting the unimodal property of Gaussian kernel, and Series-association, which is used to learn the associations from raw series, are found to be more informative. The adjacent-concentration prior and the learned real associations are also reflected in this approach and their differences are meant to be distinguishable between normal and abnormal cases [11]. Specifically, this association-based criterion attention block has demonstrated the ability to consistently yield smaller values for the normal component, which starkly contrasts the point-contextual and pattern-seasonal scenarios.

B. *Focal Loss*

Focal Loss [12] was proposed to address the problem of imbalanced data by introducing a new loss control function assigning higher weights to challenged or incorrectly classified cases. To reduce the loss contribution of highly classified examples and focus on more difficult examples is an important idea behind this loss function, shown in Eq. (1)

$$FL(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log p_t \tag{1}$$

where $FL(p_t)$ represents the focal loss for a specific example with predicted probability p_t. α_t is a balancing factor to address class imbalance. It is often defined as the inverse of the class frequency or adjusted based on the dataset characteristics. γ is the focusing parameter that controls the rate of downweighing for well-classified examples and pt represents the predicted probability for the ground truth class. By incorporating Focal Loss into the training process, the model can focus more on the challenging examples, improving its ability to handle class imbalance and boost performance, particularly in scenarios where the number of abnormal examples is significantly lower than the number of normal examples. Since faults in sensor data are expected to happen infrequently, thus, we utilize focal loss for our target model's loss function.

III. Experiment and Analysis

A. *Dataset and Evaluation Metrics*

Data-driven methods are highly prevalent and successful because they are based on data extracted from historical processes and large quantities of such data generated by modern production. However, for security reasons, this information is generally kept private. Thus, we conduct experiments using the public simulation data published by Intel Labs with injected faults from [13]. The injected fault types include bias, drift, malfunction, random, polynomial drift, and mixed, as shown in Fig. 2, along with the normal and interpolated original data. The original data is obtained from 10 sensors and a total data sample of 3,980,192. This fault detection task is evaluated using precision, recall, and the F1-score.

B. *Implementation Details*

The temperature sensor data obtained are univariate, indicating that each time series example consists of only one channel. Consequently, we convert the time series into a multivariate format with a single channel by employing a straightforward reshaping technique using the Numpy library. The model has 3 layers and is trained for 10 epochs for each phase with batch size 64 and mean squared error (MSE) as the loss function for the pretext task. This approach will enable the data to fit the model with multivariate time series input. The experiments are implemented in PyTorch with a single NVIDIA GeForce GTX 1080Ti 11GB GPU.

C. *Performance Comparison*

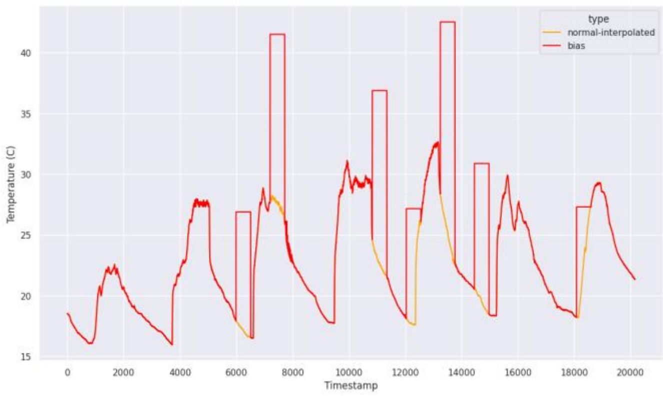
The fault detection task's performance evaluation relies on a critical metric, i.e., the probability of occurrence for the positive class containing faults. We require the commonly used performance metrics for the experiment: precision, recall, and F1-Score. Table I shows the performance comparison of our proposed method with other competing methods. FOCUSeR is based on fog computing that uses online learning and constant updating over time to avoid faults. The second method is a sensor fault diagnosis method based on α Grey Wolf Optimization Support Vector Machine (α-GWO-SVM). We compare our results with the reported results in the literature for the first two comparisons due to the code's unavailability. Lastly, the original Anomaly-Transformer is also tested on the dataset. In this experiment, we use the binary class (normal and fault) of the signal as the ground truth to the original model instead of the multi-class label. It can be seen that our method outperforms other methods on the benchmark dataset. Moreover, compared with the best performance of other methods, our method has a significant improvement on the F1-Score by 1.69% over the best results of other methods, which clearly shows that our method has good performance for the task.

TABLE I. PERFORMANCE COMPARISON OF DIFFERENT METHODS

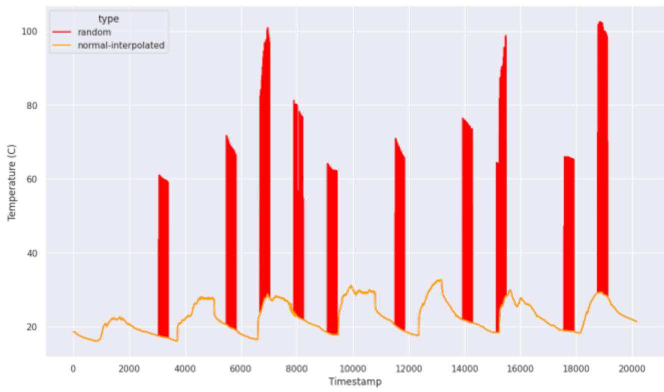
Method	Precision	Recall	F1-Score
FOCUSeR [14]	96.1%	97.7%	96.9%
α-GWO-SVM [3]	97.29%	-	-
Anomaly-Transformer [11]	90.43%	92.18%	91.29%
Ours	98.07%	99.11%	98.59%

(-), not known, the authors did not evaluate the results of their methods using certain metrics nor make their codes available.

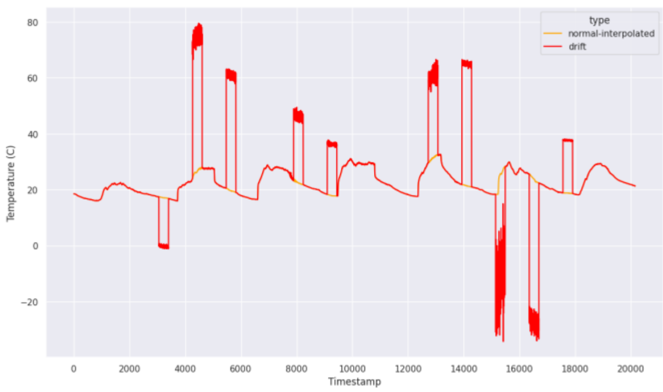
Our proposed method learns the pattern of the signals by reconstructing the clean signal from the fault-injected signal. This imposes the model to identify anomalous patterns that have never appeared in the clean signal. Moreover, the performance of the model indicates that the difference between normal and abnormal time points is effectively intensified by adapting the anomaly-attention block. Thus, our proposed method achieves the state-of-the-art result on an exhaustive set of empirical studies.



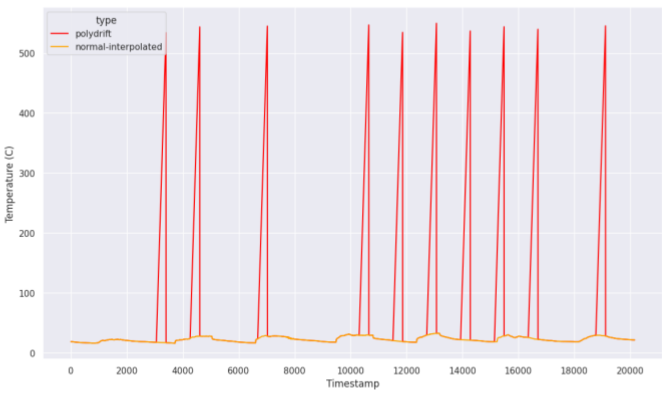
(a) Example of bias fault



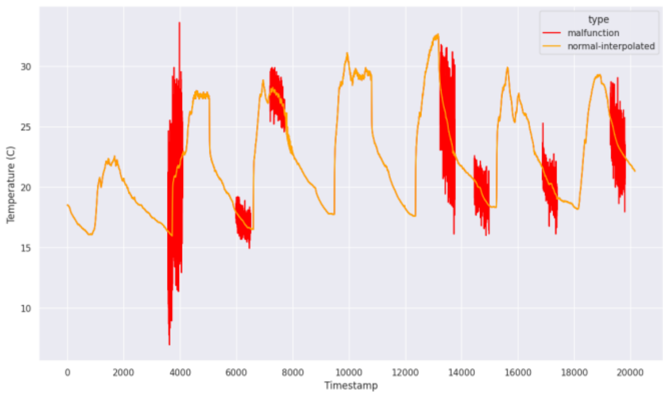
(b) Example of random fault



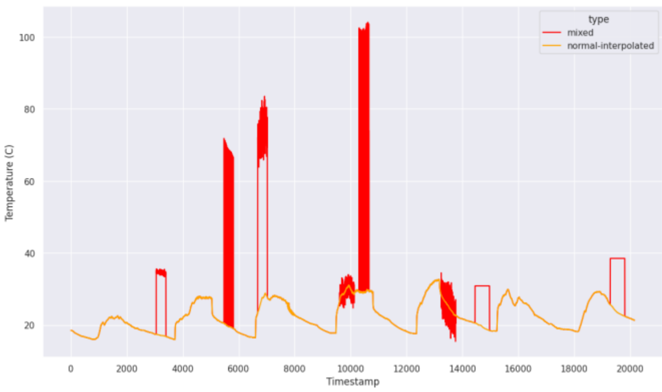
(c) Example of drift fault



(d) Example of polynomial drift fault



(e) Example of malfunction fault



(f) Example of mixed fault

Fig. 2 Examples of injected fault

From a statistical standpoint, faults are infrequent occurrences, that happen less often than the normal class. The primary way to describe this behavior is by considering the anticipated occurrence rate of anomalies. Thus, the specific advantage of applying Focal loss for the experiment is observed. Table II demonstrates the effectiveness of using Focal loss as the target model's loss function. Focal loss can differentiate various signal conditions, enabling the model to adapt more effectively. This adaptation facilitates improved fault detection accuracy by enhancing the model's ability to accurately identify faults in diverse scenarios.

TABLE II. PERFORMANCE OF PROPOSED MODEL UNDER DIFFERENT LOSS FUNCTION

Method	Loss Function	F1-Score
Pretrain	MSE	96.58%
Target	MSE	
Pretrain	FL	94.87%
Target	FL	
Pretrain	MSE	98.59%
Target	FL	

IV. CONCLUSIONS

In this research paper, we propose an effective self-supervised approach for detecting multi-fault in temperature sensor data. Firstly, we enhance the discriminative power of the anomaly attention block. Next, we incorporate attribute information of sensor data into the model training through a customized loss, enabling the detector to differentiate better and adapt to diverse sensor conditions, resulting in improved accuracy in detecting faults. Lastly, experimental results validate the effectiveness of our proposed method, and our future work involves developing a more efficient model structure for temperature sensor data multi-fault detection.

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